PhishGuard-AI Research Paper

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Abstract

Phishing is when a person send fake emails or websites to trick people into giving away their passwords or bank details. It is a big problem because many people fall for it. Most detection systems use machine learning, but they need to collect all the user data in one place. This is risky because emails and browsing history are private, and if the server gets hacked, everyone's data is exposed. So privacy matters a lot here.

In this paper, we made a system called PhishGuard-AI. It uses federated learning, which means each client trains the model on its own data and only sends the model updates to the server. The raw data never leaves the client.

We tested it on a phishing dataset that was split across many clients. We tried two setups. In the IID setup, every client has similar data. In the Non-IID setup, the data is different for each client, which is like real life.

The results were good. PhishGuard-AI got 90.24 percent accuracy for IID and 86.53 percent for Non-IID. The Non-IID result is a bit lower, but still okay.

In conclusion, we can detect phishing well without sharing private data, so PhishGuard-AI is a useful and safe way to fight phishing.

Phishing Detection, Federated Learning, Privacy-Preserving Machine Learning, Cybersecurity, Non-IID Data

1 Introduction

1.1 Background

When someone pretends to be a person or company to trick you into sharing your private information. It comes as a fake email, text, or website that looks real. For example, you might get an email that seems to be from your bank, asking you to click a link and enter your password. If you do, the attacker get all of your details.

The aftermath can be serious. People can lose money, get their accounts hacked, or have their identity stolen. In companies, just clicking a bad link can lead to a heavy data leak, which costs money, causes legal trouble. Phishing is also often how attackers start bigger attacks like ransomware. Since the fake messages keep getting more convincing, they are hard to catch, which is why we need better ways to detect them.

1.2 Problem Statement

Phishing attacks is a common cybersecurity problem, and detecting them is most important for protecting users and their information. Machine learning models are used for phishing detection, but for long time approaches require data to be collected and stored at a central location for training. This can create privacy concerns, especially when the data doesnt belongs to them but different organizations that cannot share their information.

Federated learning offers a different approach by allowing multiple clients to train a model using their own data without sharing the actual dataset. However, the data available to each client may not be the same. Some clients may have more phishing URLs, while others may mostly contain legitimate URLs. These differences can affect how well the shared model learns and performs.

The main problem addressed in this research is to study how federated learning can be used for phishing detection while keeping data decentralized. The research also investigates how different data distributions affect the model's performance and compares the results with a traditional centralized machine learning approach.

1.3 Research Objectives

The main objective of this research is to study how federated learning can be used for phishing detection while keeping data decentralized.

The specific objectives are:

1. Develop a Machine learning model for detecting phishing and legitimate URLs using data stored across multiple clients.

2. Compare IID and non-IID data distributions and learn how differences in client data affect model performance.

3. Estimate the federated model using accuracy, precision, recall, and F1-score.

4. Also compare federated learning with centralized machine learning and identify differences in detection performance.

5. To examine the benefits and limitations of federated learning for phishing detection, particularly its reduced need to share raw data.

1.4 Contributions

The key contributions of this research are:

1. Application of federated learning to phishing detection: A federated model is developed to detect phishing and legitimate URLs while keeping the training data distributed across multiple clients.

2. Evaluation under different data distributions: The study compares IID and non-IID settings to understand how differences in client data affect federated model performance.

3. Comparison with centralized learning: The federated model's performance is compared with a centralized Random Forest model using accuracy, precision, recall, and F1-score.

4. Analysis of performance trade-offs: The research examines how federated learning performs when data remains decentralized, including the challenges caused by non-IID data.

5. Experimental evaluation: The study presents performance results and comparisons to assess the potential and limitations of federated learning for phishing detection.

2 Research Methodology

2.1 Dataset Description

This research uses the dataset `dataset_B_05_2020.csv` for phishing detection. The dataset contains 11,430 URL samples, including 5,715 legitimate URLs and 5,715 phishing URLs. Each URL is represented using 87 numerical features, while the `status` column is used as the target label.

The dataset is used for training and evaluating a model that classifies URLs as either legitimate or phishing. To study federated learning, the training data is divided across multiple clients instead of being stored at a single location.

2.2 Data Preprocessing

Before training the model the URL column is removed because the model uses the numbers extracted from each URL. The target labels are converted into numerical values, where legitimate URLs are represented by 0 and phishing URLs by 1.

The dataset is divided into training and testing sets using an 80:20 split. The training set is used to train the models, while the testing set is used to evaluate their performance. Feature scaling is applied by fitting the scaler on the training data and then using it to transform both the training and testing data.

2.3 Federated Learning Setup

The federated learning system is simulated using five clients. Each client receives a portion of the training data and trains the model locally. The clients do not directly share their raw training data with the central server.

After local training, the model updates from the clients are sent to the server. The server combines these updates using Federated Averaging (FedAvg) to create a global model. The updated global model is then shared with the clients for the next training round.

2.4 IID and Non-IID Data Distribution

Two data distribution settings are considered in this research.

In the IID setting, the training samples are distributed randomly among the clients so that each client receives a similar data distribution.

In the non-IID setting, the data distribution differs across clients. Some clients may receive a higher proportion of phishing URLs, while others may receive more legitimate URLs. This setup is used to study how differences in client data affect the performance of the global model.

2.5 Model Architecture and Training

The federated model is a feed-forward neural network with an input layer containing 87 features, two hidden layers with 128 and 64 neurons, and an output layer containing two classes. ReLU activation functions are used in the hidden layers, while the output layer produces predictions for legitimate and phishing URLs.

Each client trains the model locally using Stochastic Gradient Descent (SGD) with a learning rate of 0.01. The local model updates are then combined by the server using FedAvg. The process is repeated for five communication rounds.

2.6 Performance Evaluation

The models are Tested by using accuracy, precision, recall, and F1-score. These are used to measure the model's total classification performance and its ability to identify phishing URLs.

The federated learning results under IID and non-IID settings are compared with a centralized Random Forest model trained on the same dataset. This comparison helps evaluate the performance differences between centralized and federated approaches.

3 Related Work

3.1 Machine Learning for Phishing Detection

Previous works have investigated the machine learning techniques to detect phishing URLs. These techniques use features from the URLs to classify them as a legitimate or fake . Centralized machine learning algorithms can achieve strong detection performance but usually need the training data to be collected in a central location.

3.2 Federated Learning for Cybersecurity

Federated learning has been explored as an approach for training machine learning models across multiple clients without directly sharing their raw data. This makes it relevant to cybersecurity applications where data may belong to different organizations or users.

3.3 Research Gap

Although federated learning can support decentralized training, its performance may be affected when clients have different data distributions. This research addresses this issue by comparing federated phishing detection under IID and non-IID settings and evaluating the results against a centralized machine learning baseline.

4 More Related Work

4.1 Machine Learning for Phishing Detection

Machine learning has been widely studied for phishing website and URL detection. Safi and Singh reviewed several approaches, including blacklist-based, heuristic, machine learning, and deep learning techniques. Their review highlights the use of machine learning algorithms, particularly Random Forest, in phishing detection research [1]. These approaches can identify phishing patterns using features extracted from URLs and websites. However, many traditional methods rely on training data being collected at a central location.

4.2 Federated Learning for Phishing Detection

Federated learning provides an alternative by allowing multiple clients to train a shared model without directly transferring their raw data to a central server. Sakazi et al. proposed a semi-supervised, transfer-learning, and federated-learning approach for phishing URL detection. Their method uses decentralized data and aggregates model updates through Federated Averaging [2].

This work demonstrates the potential of federated learning for phishing detection. However, the distribution of data across clients can vary, and this may influence the performance of the global model.

4.3 Research Gap

Existing research has explored both centralized machine learning and federated learning for phishing detection. However, understanding how differences in client data distribution affect model performance remains important.

This research mainly on evaluating a federated phishing detection model under IID and non-IID data distributions. Results are compared with a Random Forest baseline using accuracy, precision, recall, and F1-score. The aim

is to study the performance trade-offs of federated learning when training data remains distributed across clients.

References

[1] A. Safi and S. Singh, "A systematic literature review on phishing website detection techniques," *Journal of King Saud University - Computer and Information Sciences*, vol. 35, no. 2, pp. 590–611, 2023, doi: 10.1016/j.jksuci.2023.01.004.

[2] I. Sakazi, E. Grolman, Y. Elovici, and A. Shabtai, "Utilizing a semi-supervised, transfer-learning, federated-learning approach to detect phishing URL attacks," in *Proc. International Joint Conference on Neural Networks (IJCNN)*, 2024, doi: 10.1109/IJCNN60899.2024.10650184.

5 Results and Discussion

5.1 Performance Comparison

Table 1 presents the performance of the centralized Random Forest model and the federated model under IID and non-IID data distributions.

Table 1: Performance Comparison of Phishing Detection Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Centralized Random Forest	96.06%	95.74%	96.41%	96.08%
Federated IID	90.24%	89.66%	90.99%	90.32%
Federated Non-IID	86.53%	92.13%	79.88%	85.57%

5.2 Discussion

The centralized Random Forest model achieved the highest accuracy and F1-score among the evaluated models. The federated model under the IID setting performed better than under the non-IID setting, suggesting that differences in client data distribution can make federated training more challenging.

The non-IID model achieved higher precision than the IID model but lower recall. This indicates that it was more selective when predicting phishing URLs, but it missed a larger number of actual phishing samples. These results show that client data distribution affects the balance between detecting phishing URLs and avoiding false alarms.

5.3 Graphical Analysis

The graphical results provide a visual comparison of the evaluated phishing detection models. Figure 1 shows that the centralized Random Forest model

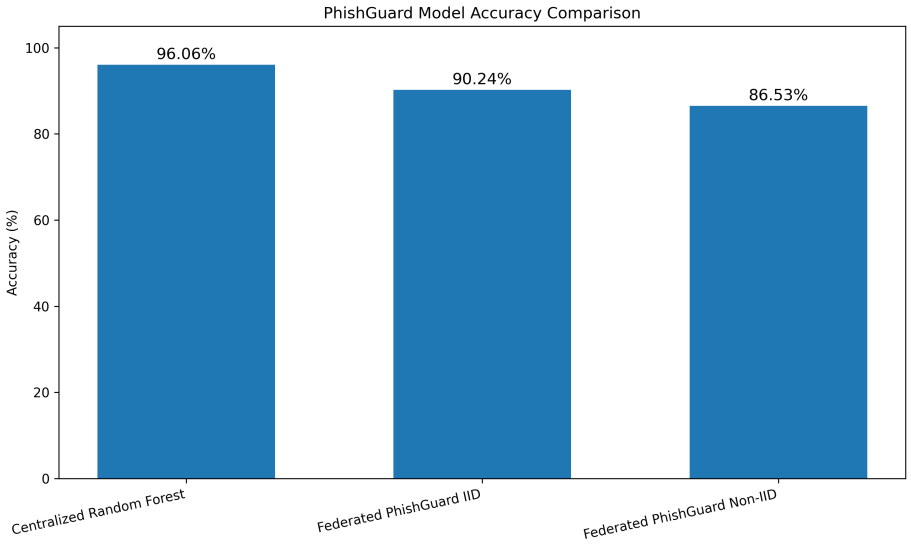


Figure 1: Accuracy comparison between centralized and federated phishing detection models.

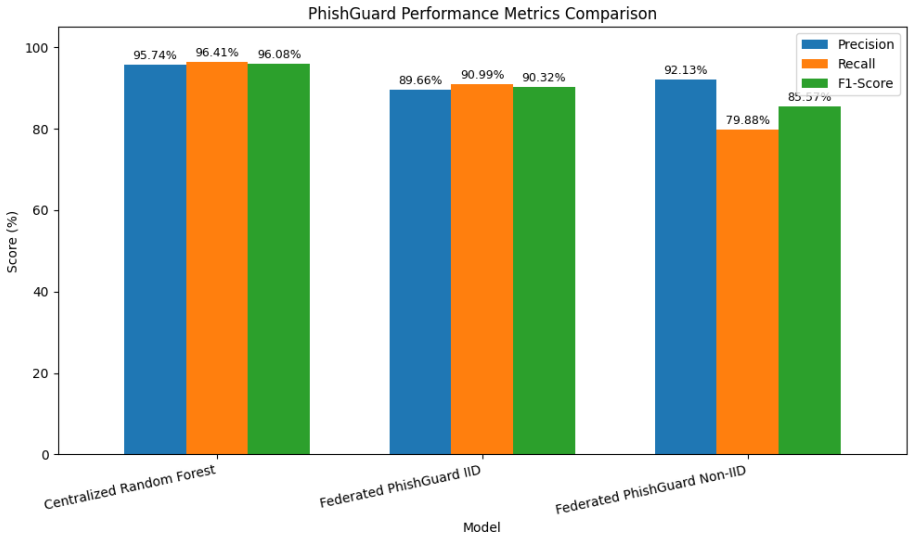


Figure 2: Comparison of performance metrics across the evaluated models.

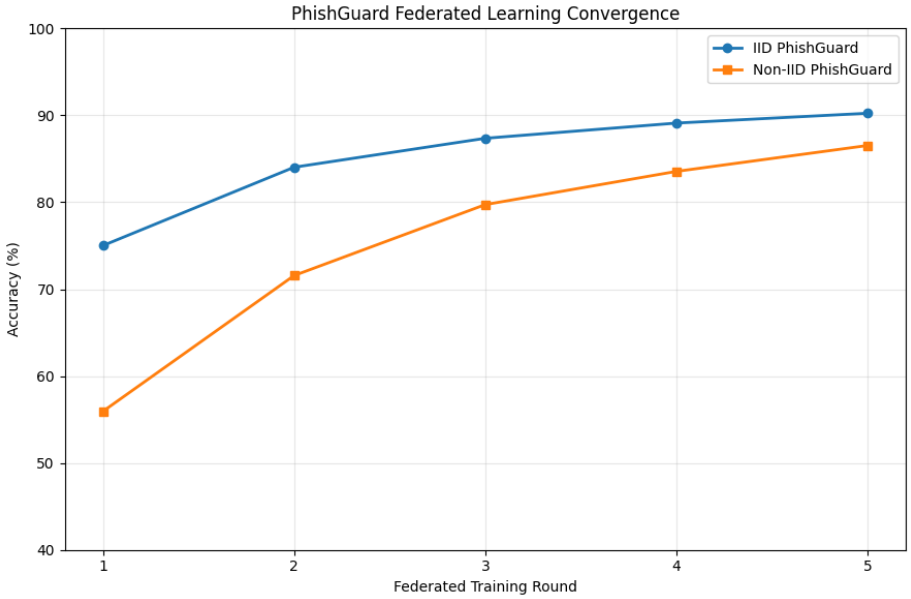


Figure 3: Federated model performance under IID and non-IID data distributions.

achieved the highest accuracy at 96.06%, followed by the federated IID model at 90.24% and the federated non-IID model at 86.53%.

Figure 2 compares precision, recall, and F1-score. The non-IID model achieved higher precision than the IID model, but its recall was lower. This indicates that the non-IID model identified fewer false phishing alerts but missed more actual phishing URLs.

Figure 3 shows the accuracy of the federated model over five communication rounds. Both IID and non-IID models improved as training progressed. The IID model have higher accuracy in each round, while the non-IID model showed steady result. These results suggest that differences in client data distribution can affect federated learning performance.

6 Conclusion

This research studied the use of federated learning for phishing detection by comparing model performance under IID and non-IID data distributions. A federated neural network was trained across multiple clients, and its performance was compared with a centralized Random Forest model.

The experiments show that the data distribution across clients affects federated model performance. The IID setting achieved better results than the non-IID setting, indicating that differences in client data can make federated

training more challenging. The centralized Random Forest model also achieved higher performance than the federated model in the current experiments.

Overall, the results show that federated learning can be used for phishing detection while also keeping training data distributed across clients. However, further work is needed to improve performance under non-IID conditions and evaluate additional privacy mechanisms. Future work may include sample-weighted aggregation, more communication rounds, and techniques such as differential privacy or secure aggregation.